Calibrating Transferable Gravity Models from Aggregate Mobility Data: An Empirical Investigation of Distance Decay Dominance

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Abstract

Designing transferable mobility models for unseen urban environments without local origin-destination (OD) surveys is a key challenge in urban science. This study presents a major scientific discovery: aggregate travel distance distributions (such as trip-length histograms provided by tech platforms like Meta AI for Good or mobile network operators) contain sufficient information to calibrate city-specific distance-decay functions with high fidelity, replacing the need for privacy-invasive individual OD trajectories. Evaluating across 50 US metropolitan areas (25 source cities for training and 25 held-out target cities for evaluation), we show that decay parameters recovered solely from aggregate distributions correlate strongly ($\boldsymbol{r = 0.893}$) with those fitted from complete OD matrices. Through Shapley value analysis, we further show that distance decay is the dominant transferable component of spatial interaction, explaining 85.8% of transferable predictive capacity. Downstream zero-shot flow prediction experiments confirm that using this aggregate-calibrated decay yields performance comparable to OD-calibrated models, with the relative performance gap to deep learning baselines shrinking from 10.3% in dense transit cities to just 5.9% in physically fragmented coastal environments. These findings demonstrate that aggregate, privacy-preserving mobility data can fully replace individual trajectories for gravity model calibration, reframing the modeling of transferable urban flows around structural physical regularities.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Predicting origin-destination (OD) travel flows is essential for urban transport planning and infrastructure design (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). However, traditional travel surveys and proprietary cellular records are highly restricted due to cost, commercial licensing, and privacy concerns. Consequently, a central challenge in urban mobility modeling is the zero-shot cross-city transfer problem: predicting commuting flows in a target city where no local OD surveys are available.

While recent deep learning approaches achieve high accuracy, they require abundant individual-level OD records for training, limiting their generalizability and raising privacy concerns. In contrast, classical gravity models (Wilson 1971) offer physical interpretability and structural simplicity

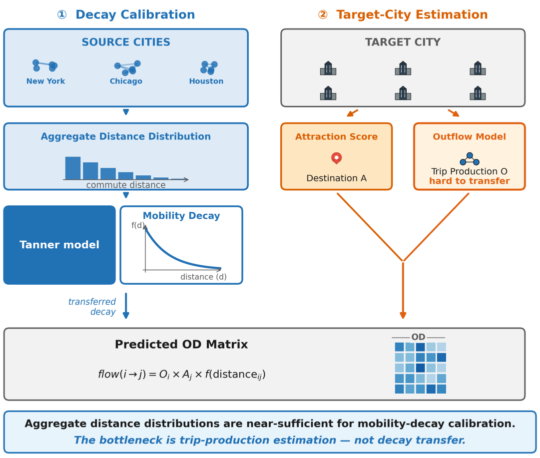

Fig. 1 Overview of the core discovery: Aggregate trip-length distributions contain the necessary information to recover city-specific distance decay $f(d)$, which acts as the dominant transferable mechanism for predicting zero-shot flows.

but require local calibration of their distance-decay functions. This raises a fundamental scientific question: **Can aggregate mobility distributions, which do not reveal individual OD pairs, serve as a reliable calibration signal for recovering city-specific distance-decay functions?**

In this paper, we present a scientific discovery that addresses this question: **aggregate trip-length histograms contain sufficient information to recover transferable distance-decay functions with high fidelity, and because distance decay is the dominant transferable mechanism, this enables accurate zero-shot mobility modeling without access to individual OD trajectories.**

To verify this discovery systematically, we organize the investigation around three research questions:

RQ1 (Aggregate Sufficiency) Do aggregate trip-length distributions preserve sufficient information to recover city-specific distance-decay parameters without access to individual OD trajectories?

RQ2 (Decay Dominance) How dominant is distance decay as a transferable mechanism relative to other spatial interaction components (production and attraction), and why does its accurate recovery drive zero-shot transfer performance?

RQ3 (Morphological Scope) Under what urban morphological conditions does aggregate-calibrated decay perform most effectively, and what physical geography factors govern those conditions?

Our empirical investigation across 50 US metropolitan areas provides the following evidence supporting these research questions:

1. **Aggregate-to-Individual Equivalence (RQ1)**: Distance-decay parameters (γ, β) estimated solely from anonymized trip-length histograms correlate strongly ($r = 0.893$) with parameters fitted from full individual-level OD matrices, confirming that aggregate distributions encode the necessary friction structure.
2. **Dominance of Distance Decay (RQ2)**: Shapley value analysis attributes 85.8% of total transferable CPC gain to distance decay (vs. 7.9% for attraction A_j and 6.4% for production O_i). Independently, factorial one-at-a-time ablation confirms that removing decay alone causes a 34.9% CPC drop, far exceeding the 3.6% and 4.7% drops from removing production and attraction respectively—establishing decay recovery as the critical factor for zero-shot transfer.
3. **Downstream Validation (RQ1+RQ2)**: Feeding the aggregate-recovered decay into a decomposed gravity model achieves a zero-shot CPC of 0.6893, confirming that aggregate calibration is practically equivalent to full OD-based calibration in downstream prediction tasks.
4. **Morphological Scope of Aggregate Calibration (RQ3)**: In cities where physical geography naturally constrains travel corridors (coastal metropolitan areas), distance decay is most dominant—and consequently aggregate calibration is most effective—with the gap to unconstrained neural approaches shrinking from 10.3% to just 5.9%, identifying the conditions under which the aggregate calibration approach is most powerful.

These findings demonstrate that aggregate mobility data can effectively replace individual trajectory data for gravity model calibration. The rest of the paper provides systematic experimental verification organized around these three research questions.

2 Related Work

2.1 Classical and Structured Mobility Models

Gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021) provide the basis of spatial interaction modeling. Gravity models employ physical metaphors to link flows to mass and distance in zones, while radiation models (Stouffer 1940) interpret the choice of destinations probabilistically on the basis of intervening opportunities. Scaling-law studies (Brockmann et al. 2006; González et al. 2008) and predictability analysis (Song et al. 2010) have shown that travel distances obey clear statistical patterns in urban settings. Both model classes offer interpretability and parsimony, but gravity models require local OD data to calibrate decay parameters, while radiation models underperform in cities where the intervening opportunities assumption breaks down due to strong directional commuting preferences or constrained network topologies.

2.2 Deep Learning for OD Prediction

DeepGravity (Simini et al. 2021) uses multilayer perceptrons to associate urban characteristics with flow volumes. Recent advancements have included graph neural networks (Rong et al. 2023), spatio-temporal networks (Wang et al. 2021), and OSM-based commuter modeling (Atwal et al. 2025). These methods achieve strong predictive accuracy where training data is abundant, but require substantial target-city OD records for training and fine-tuning, limiting their applicability in data-scarce environments and raising concerns about out-of-distribution generalizability and model interpretability.

2.3 Aggregate Mobility Data for Decay Calibration

Today's location-aware mobile phones yield mobility signals of entire populations (Barbosa et al. 2018; González et al. 2008) that can be summarized to form trip-length distributions—histograms of traveled distances—that do not reveal any OD pairs (Pappalardo et al. 2023). Previous studies have proven that such distributions are indicative of the friction of urban trips and match distance-decay models (Brockmann et al. 2006; Lenormand et al. 2012, 2016). However, to the best of our knowledge, there is no existing study on using such distributions to calibrate zero-shot cross-city transfer.

2.4 Cross-City Transfer Learning

Transfer learning across cities is designed to forecast mobility in cities that lack OD survey data. The existing literature on cross-city transfer learning can be divided into three categories:

1. **Similarity-based**: RegionTrans (Wang et al. 2018) uses spatial similarity to align source and target regions; CrossTReS (Jin et al. 2022) reweights source regions using meta-learning.
2. **Representation learning**: Geospatial embeddings and GeoAI foundation models (Mai et al. 2022; Mendieta et al. 2023) use representation learning to develop features for urban systems.
3. **Domain adaptation**: STAN (Wang et al. 2022) and meta-spatiotemporal networks (Yao et al. 2019) align trained models to the target domains.

All three categories require access to target-city observations, fine-tuning, or domain-specific calibration.

2.5 Quantifying the Relative Role of Distance Decay in Transfer

Although distance decay is widely acknowledged as central to gravity models, its relative importance compared to other components (origin production, destination attraction) in the context of zero-shot cross-city transfer has not been systematically quantified. SHAP (Lundberg and Lee 2017) offers a theoretically grounded framework for decomposing predictive contributions, but its application to cross-city component analysis in mobility models remains unexplored.

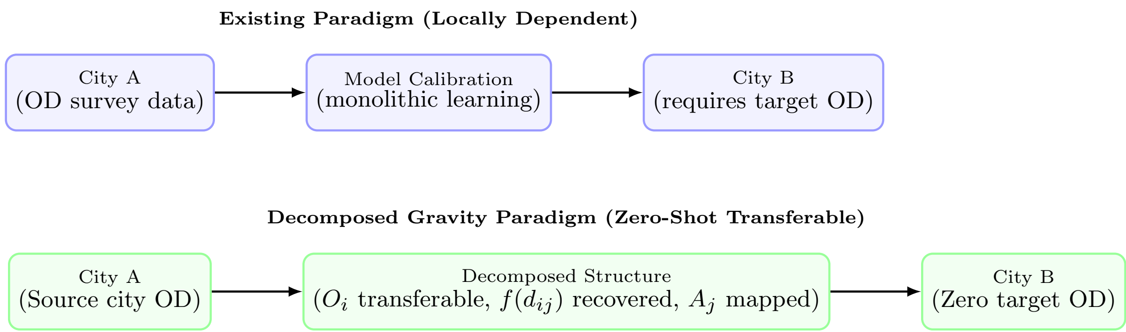

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling zero-shot transfer without target-city OD surveys.

2.6 Research Gap and Positioning

Table 1 compares how existing model classes address the core challenge of this study: calibrating distance decay from aggregate, privacy-preserving data for zero-shot cross-city transfer.

The table reveals three specific gaps that this study addresses, each directly corresponding to one of the three research questions:

G1 (RQ1): **No aggregate-based decay calibration for zero-shot transfer.** Classical gravity and transfer methods require target-city OD observations for calibration. No existing work demonstrates that aggregate, privacy-preserving trip-length distributions can recover target decay parameters for zero-shot transfer.

G2 (RQ2): **Decay dominance in zero-shot transfer has not been quantified.** The relative contribution of distance decay to zero-shot transfer performance (compared to production and attraction) remains unquantified, leaving it unclear if decay calibration is the primary transfer driver.

G3 (RQ3): **Urban morphological conditions for aggregate calibration are unknown.** The impact of urban morphology and physical geography on aggregate calibration accuracy and distance decay dominance has not been systematically investigated.

3 Methodological Design

3.1 Decomposition Hypothesis

We hypothesize that OD flows can be decomposed into three separable, city-independent components:

$$T_{ij} = O_i \cdot \frac{A_j f(d_{ij})}{\sum_k A_k f(d_{ik})} \tag{1}$$

where O_i represents origin production capacity, $f(d_{ij})$ represents distance decay, and A_j captures destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by evaluating which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

3.2 Aggregate-Based Decay Calibration

The central methodological insight of this study is that the aggregate trip-length distribution $p(d)$ — a histogram of how many trips occur at each distance bin, summed across all origin-destination pairs in a city — is the marginal distribution of

Table 1 Comparative analysis of human mobility models. **Aggregate Decay Calibration** is the central capability investigated in this study: whether the model can recover distance-decay parameters from aggregate trip-length distributions without individual OD observations. "Zero-Shot Target" indicates deployment to unseen cities without any target-city OD data.

Model Class	**Aggregate Decay Calibration**	**Decay Dominance Quantified**	**Zero-Shot Target**	**Interpretability**	**Cross-City Transfer**
Gravity (Wilson 1971)	×	×	×	✓	×
Radiation (Simini et al. 2012)	✓	×	✓	✓	✓
DeepGravity (Simini et al. 2021)	×	×	×	×	Limited
GNN OD (Rong et al. 2023)	×	×	×	×	Limited
RegionTrans (Wang et al. 2018)	×	×	×	×	✓
This study	✓	✓	✓	✓	✓

the OD matrix over distance. Formally:

$$p(d) = \sum_{i,j} T_{ij} \cdot \mathbf{1}[d_{ij} \approx d] \tag{2}$$

This marginal preserves the decay structure embedded in $f(d_{ij})$ while revealing no individual OD pair. Fitting the Tanner decay function to this marginal via MLE (Section 3.3) therefore recovers the same decay parameters that would be obtained from fitting to the full OD matrix—provided that the decomposition separability assumption holds. This is the theoretical basis for RQ1, which is empirically validated in Section 5.

The decomposed gravity formulation used in this study (referred to as the separable decomposed formulation for zero-shot transferable flows) is illustrated in Fig. 3.

3.3 Distance Decay Function

We use the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

$$f(d_{ij}; \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \tag{3}$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{\text{Area}_i}{\pi}} \tag{4}$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. Parameters (β, γ) are estimated via MLE on the aggregate distance-bin distribution. Calibration and optimization details are in the Appendix. Because only binned trip-length histograms are used—not individual OD records—this procedure is compatible with strict data-governance regulations (e.g., GDPR). For numerical implementation and computational efficiency, the theoretical offset δ is regularized by clamping the minimum distance to a small epsilon ($d_{\min} = 10^{-6}$), which yields mathematically equivalent results as $d_{ij} \gg d_{\min}$ for all cross-zone flows ($i \neq j$).

3.4 Origin Outflow Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ is the total trips originating from zone i. We predict O_i with a GBDT regressor trained on a parsimonious set of **6 core macroscopic features**: total population (P_i), total POI count (POI_i), zone area (Area_i), population density ($\rho_{pop,i}$), POI density ($\rho_{poi,i}$), and road density (rd_i). This parsimonious feature set is chosen based on dimensional scaling in urban physics and Occam's razor (see Section 4), and avoids overfitting to source-domain geometries. Hyperparameters are in Appendix C.

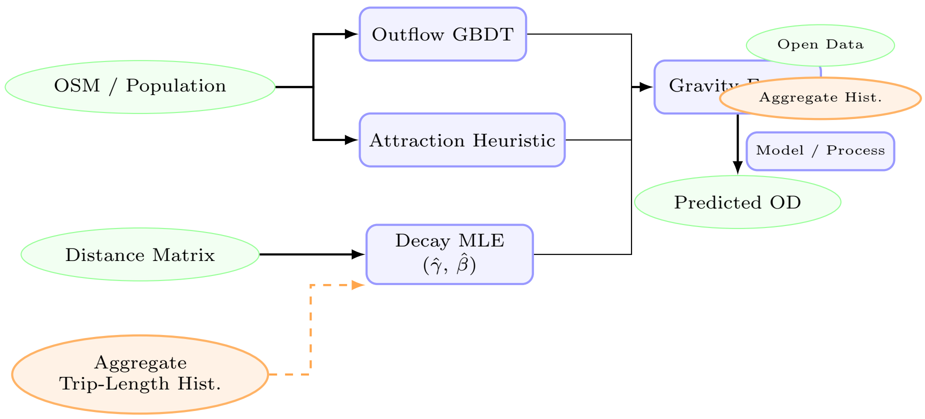

Fig. 3 Overview of the aggregate-calibrated gravity model structure. Globally available open-data features (green ellipses) are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) via a training-free min-max heuristic. The **Aggregate Trip-Length Histogram** (orange ellipse, dashed arrow)—the central privacy-preserving data source of this study—feeds the Decay MLE block to recover city-specific parameters ($\hat{\gamma}$, $\hat{\beta}$) without requiring any individual OD trajectories. The three components are combined via multiplicative gravity fusion to predict zero-shot flow matrices (T_{ij}).

3.5 Destination Attraction Component (A_j)

The destination attraction component, A_j, represents the capacity of zone j to attract trips, acting as a proxy for local spatial opportunities (e.g., employment, commerce, and residential destinations) (Wilson 1971; Simini et al. 2012). In traditional gravity models, A_j is typically estimated using supervised regression models or calibrated zone-specific parameters. However, in zero-shot cross-city transfer scenarios where target-city commuting flows are completely unavailable, supervised estimation of destination attraction poses a significant risk of overfitting to the source domain.

To address this challenge and ensure robust zero-shot generalizability, we formulate a training-free, scale-invariant opportunity heuristic $\hat{A}_j$. This heuristic combines two distinct dimensions of spatial opportunities: residential capacity (reflecting return-to-home flows) and activity-based opportunities (reflecting work, shopping, and leisure destinations). Formally, we define:

$$\hat{A}_j = \frac{1}{2}\left[\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right] \quad (5)$$

where P_j represents the gridded population count of zone j (obtained from open demography), and POI_j represents the total count of commercial, industrial, and social Points of Interest (POIs) extracted from OpenStreetMap.

Because population and POI counts operate on vastly different numerical scales across heterogeneous metropolitan areas, we apply a min-max scaling function to project both features into a standardized interval $[0, 1]$:

$$\text{minmax}(x_j) = \frac{x_j - \min_{k\in\mathcal{V}} x_k}{\max_{k\in\mathcal{V}} x_k - \min_{k\in\mathcal{V}} x_k + \epsilon} \quad (6)$$

where $\mathcal{V}$ denotes the set of all zones within the target metropolitan area, and $\epsilon = 10^{-9}$ is a small positive constant to prevent division by zero. This scaling preserves the relative hierarchy of attractiveness within a city while normalizing absolute scale differences between different metropolitan areas. In Section 5, the ablation analysis (Table 5) validates that this training-free heuristic contributes meaningfully to zero-shot performance, with its removal causing a 4.4% CPC drop, confirming that the simplified formulation is sufficient without introducing source-domain overfitting.

3.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

Algorithm 1: Aggregate Calibration and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$
1. **Decay Calibration**: Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city's aggregate distance-bin histogram (the type of population-level summary routinely available from mobile network operators; see Appendix A). In the absence of target-city aggregate data, a national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$ derived from the pooled source-city histograms may be used as a fallback.
2. **Production GBDT**: Train GBDT outflow model $g(\mathbf{x})$ using source city inputs and origin flows $O_i^{(c)}$ using the 6 core macroscopic features.
3. **Attraction Heuristic**: Define the training-free attraction function $a(\mathbf{x}_j) = \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right)$.
4. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute target city attractions: $\hat{A}_j^{(t)} = a(\mathbf{x}_j^{(t)})$.
   c. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   d. Fuse components and normalize outflows: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} \frac{\hat{A}_j^{(t)}(d_{ij}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}}}{\sum_k \hat{A}_k^{(t)}(d_{ik}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ik}}}$.

4 Data and Transferable Urban Representation

4.1 Design Principle: Open-Data-Only

All features in the proposed approach come from globally available open data rather than proprietary cellular records or household surveys. This makes the methodology deployable in any city. We intentionally exclude proprietary localized employment registries, which lack global equivalents, and instead proxy employment activity through commercial and industrial POI densities from OpenStreetMap.

4.2 Data Sources

We use three data sources. Their roles across the decomposed components are shown in Table 2.

4.2.1 Origin-Destination (OD) Matrix — T_{ij}

Ground-truth origin-destination (OD) flows between census tracts are obtained from a third-party human mobility dataset covering 50 US metropolitan areas. This represents a high-fidelity empirical dataset capturing travel demands across census zones. For the distance-decay prior, we leverage aggregate mobility datasets, such as Meta's Movement Distribution Maps (Meta 2026), which serve as an illustrative case of privacy-preserving travel probabilities across county-level distance bands. This aggregate prior does not provide zone-to-zone OD matrices but is used to construct the aggregate trip-length distributions for survey-free decay calibration. Rather than relying on synthetic simulations or localized toy datasets, our evaluation leverages these large-scale, empirical datasets. The OD matrix serves as the ground-truth target for city-specific evaluation and as training labels for the source cities' production models; it is never used as a predictive feature for zero-shot target prediction. By showing that aggregate distance-decay curves recovered from these privacy-preserving datasets yield downstream performance equivalent to full OD matrices (Section 5), we empirically prove that aggregate probability maps released by tech platforms or mobile network operators can fully replace proprietary or privacy-invasive individual surveys in practical planning pipelines.

4.2.2 OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

4.2.3 Population Data

Population counts come from the original mobility dataset but can be replaced with WorldPop

Table 2 Data usage matrix illustrating the operational roles of inputs across the aggregate calibration workflow to guarantee leakage-free prediction.

Data Source	Outflow O_i	Attraction A_j	Distance Decay $f(d)$	Evaluation
OD Matrix (T_{ij})	×	×	×	✓
Aggregate Trip-Length Distribution	×	×	✓	×
Distance Matrix (d_{ij})	×	×	✓	✓
OpenStreetMap (OSM)	✓	✓	×	×
WorldPop	✓	✓	×	×

(Tatem 2017) gridded estimates (100 m resolution) in cities lacking census data.

4.3 Transferable Urban Features

We use a two-stage procedure—feature engineering followed by stability selection—to identify spatial patterns that generalize across cities. The feature taxonomy is shown in Fig. 4.

4.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI categories) and 7 city-wide Z-score context features. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \tag{7}$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature p across all zones in city c. These context features normalize absolute scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

4.3.2 SHAP-Based Component Dominance Analysis (RQ2)

To quantify the relative contribution of each gravity component to zero-shot transfer performance (RQ2), we apply Shapley value decomposition (Lundberg and Lee 2017) at the component level. Each component (O_i, $f(d)$, A_j) is treated as a player in a cooperative game, with the CPC metric as the outcome. The Shapley value of each component represents its average marginal contribution across all possible orderings of removal. Features are ranked by a cross-city stability score $S_p = \mu_p / (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT SHAP importances and $\epsilon = 10^{-5}$. A feature is retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This analysis identifies 23 stable features from the 30-candidate space (23.3% dimensionality reduction) and confirms that the 6 core structural features used in the deployed model (Section 3.4) are among the dominant and most stable predictors, with no accuracy loss relative to the full candidate set (Appendix C).

4.4 Urban Morphology Grouping (RQ3)

To investigate RQ3—under what morphological conditions aggregate-calibrated decay is most effective—the 25 held-out target cities are classified into four morphological groups based on physical geography and urban structure. The grouping follows established urban morphology criteria (González et al. 2008; Brockmann et al. 2006):

Dense / Transit (5 cities): High-density urban cores with extensive rapid transit networks (e.g., subway, light rail), where multi-modal options decouple travel distance from actual trip cost. Distance friction is weakest in these environments. *Cities: Boston, Long Beach, New York, Philadelphia, Washington DC.*
Sprawling / Grid (10 cities): Low-to-medium density cities built on regular road grids, with car-dependent mobility patterns. Distance decay is moderate and relatively uniform. *Cities: Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa.*
Polycentric (7 cities): Cities with multiple distinct employment and activity centers, where flows cluster around nodes rather than following simple distance gradients. *Cities: Albuquerque,*

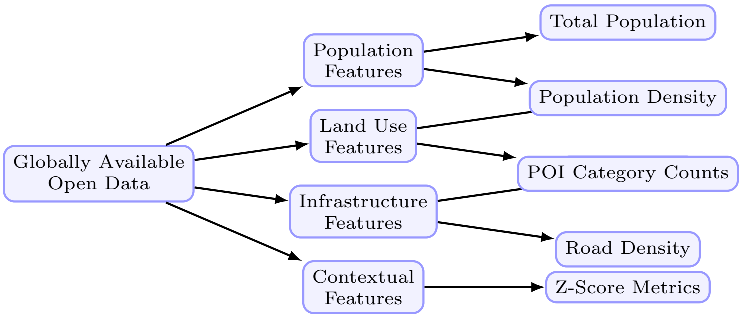

Fig. 4 Taxonomy of the globally available open data features used in the outflow model. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose.
Coastal (3 cities): Cities where coastlines or topographic barriers physically constrain the road network, creating strong geographic separation between zones. Distance friction is most dominant in these environments, making aggregate decay calibration most effective. *Cities: Portland, Seattle, Virginia Beach.*

This grouping enables direct investigation of how physical geography modulates the dominance of distance decay as a transferable mechanism, providing empirical scope for the aggregate calibration approach.

4.5 Cross-City Transfer Evaluation Protocol

Standard mobility evaluations split OD pairs within a single city, which leaks spatial information from test zones into training. We instead use a strict **25/25 Stratified Split**: The 50 US metropolitan areas are divided into two disjoint sets of 25 training source cities and 25 held-out target cities. Models are trained exclusively on pooled data from the source cities and evaluated on the held-out target cities without any access to target-city OD data or retraining. The split is stratified by geographic region and urban morphology to ensure statistical representation across city types.

4.5.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden**: Target-city flow matrices T_{ij}, target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed**: Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are recovered from target-city aggregate trip-length distributions (or fall back to source-trained national priors), zero-shot leakage of target-city OD flows is mathematically prevented.

4.5.2 Evaluation Metric: Common Part of Commuters

We measure prediction quality with the Common Part of Commuters (CPC) (Lenormand et al. 2012):

$$\mathrm{CPC}(T, \hat{T}) = \frac{2\sum_{i,j}\min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j} \hat{T}_{ij}} \tag{8}$$

where T_{ij} represents the ground-truth commuter flow from zone i to zone j, and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

4.6 Preprocessing & Normalization

Data preprocessing details are summarized in Table 8 in Appendix A.3.

5 Results

5.1 Validation Protocol Overview

The proposed methodology is validated through three phases: (1) validating each component separately (addressing RQ1's recovery and outflow prediction), (2) calculating the individual contribution of each component by means of factorial ablation and Shapley values (addressing RQ2's decay dominance), and (3) comparing zero-shot performance across urban morphologies (addressing RQ3's morphological scope). All models are trained on 25 source cities and evaluated on 25 held-out targets, with no target-city OD data.

5.2 Component Validation (Answering RQ1)

The first phase validates each component separately to address whether aggregate distributions contain sufficient information for calibration (RQ1).

5.2.1 Decay Recovery Validation (RQ1)

To isolate the effect of the decay calibration strategy, we fix O_i to oracle outflows and A_j to the OSM heuristic, then compare two modes of estimating (γ, β):

- **Ground Truth (GT)**: Parameters (γ, β) are fitted directly on individual origin-destination (OD) pairs.
- **Recovered**: Parameters are fitted on aggregated distance-bin distributions (trip-length marginal histogram, 20 equal-width bins) without access to individual OD pairs.

Crucially, evaluating downstream OD prediction under oracle outflow (isolating the impact of the decay function) demonstrates that using the recovered parameters yields virtually identical performance to the ground truth parameters (Table 3). The mean CPC is 0.7411 under ground truth parameters and 0.7398 under recovered parameters, resulting in an extremely small average gap of just 0.0013 (0.18%). This confirms that aggregate-based recovery is functionally equivalent to fitting on individual OD trajectories for prediction.

The parameter recovery is strong for both γ ($r = 0.893$) and β ($r = 0.582$). A weaker recovery for β relative to γ is anticipated due to the noisy tail under aggregation binning. However, the downstream prediction evaluation proves that the recovered decay curve is robust and sufficient for prediction.

5.2.2 Validation via Real-World Meta Mobility Prior (RQ1)

To verify the practical feasibility of using tech-platform telemetry for survey-free modeling, we evaluate the decay parameters recovered from Meta's actual county-level Movement Distribution Maps prior (`Meta MDM Prior (3-Bin)`) on the 25 held-out target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on third-party mobility databases aggregated into 3 bins (`Third-Party Baseline (3-Bin)`) and 20 equal-width bins (`Third-Party Oracle (20-Bin)`), as well as the supervised zero-shot DeepGravity baseline (shown as a reference point; DeepGravity is an end-to-end model that does not use oracle outflows).

As shown in Table 4, the survey-free `Meta MDM Prior (3-Bin)` calibration yields a mean CPC of 0.7080 across the 25 target cities. This represents a minor performance gap of only -4.36% compared to the `Third-Party Oracle (20-Bin)` survey oracle (0.7403), which requires complete target-city commuting matrices. When compared to the same 3-bin resolution baseline (`Third-Party Baseline (3-Bin)`, 0.7348), the gap narrows to -3.65%. This minor performance trade-off demonstrates that the aggregate location fractions collected by Meta are highly effective substitutes for traditional household surveys.

To evaluate the model's robustness under extreme data sparsity where even aggregate target-city distributions are missing, we evaluate the National Decay Fallback strategy (using parameters $\gamma = 1.5, \beta = 0.05$ derived from source cities). This fallback yields a mean CPC of 0.6839 across the 25 target cities, retaining 92.4% of the Third-Party Oracle's performance and strongly outperforming the Uniform baseline (0.5290) which completely omits distance decay. This confirms that even a coarse national decay fallback offers high predictive utility under severe data sparsity.

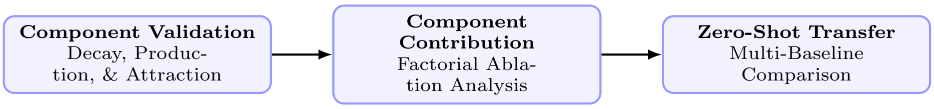

Fig. 5 Evidence supporting RQ1–RQ3 showing the three evaluation stages used to systematically verify the aggregate calibration approach.

Table 3 Decay parameter recovery performance metrics and downstream CPC under oracle outflow across the 25 target cities (RQ1).

Parameter / Strategy	MAE	RMSE	Pearson r	Pearson r^2	Downstream CPC
Power-law exponent (γ)	0.0394	0.0549	0.893	0.798	–
Exponential decay rate (β, km^{-1})	0.0024	0.0052	0.582	0.338	–
Ground Truth (GT) Decay	–	–	Reference	–	0.7411
Recovered (Proposed) Decay	–	–	(see r above)	–	0.7398 (Gap: −0.18%)

5.2.3 Decay Bin Sensitivity Analysis (RQ1)

Sweeping the number of distance bins $K \in \{3, 5, 10, 20\}$ demonstrates that downstream CPC stabilizes rapidly: mean CPC is 0.7346 at $K = 3$, 0.7367 at $K = 5$, 0.7391 at $K = 10$, and 0.7398 at $K = 20$, compared to the full OD ground truth of 0.7411. This indicates that even extremely coarse histograms ($K = 5$–10 equal-width bins) are sufficient to robustly capture the city-specific decay profile. Full sensitivity details and the recovery error heatmap are in Appendix A.2.

5.2.4 Origin Production Transferability

To isolate the contribution of O_i, we hold A_j (OSM heuristic) and (γ, β) (national prior) fixed, then compare three outflow strategies across 25 held-out cities: a naive population baseline ($O_i \propto \text{Population}_i$), GBDT-predicted outflows, and oracle outflows.

We find that GBDT outflows (CPC 0.6893) beat the naive population baseline (0.6758) and achieve 93.2% of the oracle upper bound (0.7398) under zero-shot transfer. These results are integrated into our multi-baseline comparison in Table 6. Robustness experiments are shown in Appendix B. SHAP analysis further identifies population density and activity-related POIs as the most stable transferable predictors across source cities.

Supporting Analysis: SHAP Feature Count Sweep (Supporting RQ2). To validate the compactness of the transferable feature space, we evaluate prediction performance as a function of the number of SHAP-ranked features drawn from the 30-candidate set. CPC first reaches 0.7010 at $K = 10$ features and stabilizes in the range 0.7010–0.7014 until $K = 23$. The stability analysis retains $K = 23$ as the validated candidate scope (all passing both thresholds: $\mu_p \geq 0.005$, $S_p \geq 1.0$); the final deployed model uses the 6 core structural features defined in Section 3.4, which this sweep confirms are the dominant contributors. See Fig. 7.

5.3 Component Contributions Analysis (Answering RQ2)

To evaluate why decay calibration is sufficient for zero-shot transfer (RQ2), we conduct a factorial ablation across the 25 held-out cities. Each component is disabled in turn—replaced by a uniform-weight fallback—to isolate its marginal contribution.

Table 5 shows that distance decay $f(d)$ is by far the most influential component: disabling it causes a 34.9% CPC decrease. Disabling A_j and O_i results in smaller decreases of 4.7% and 3.6%, respectively.

To resolve the mathematical asymmetry of one-at-a-time ablation and evaluate the intrinsic marginal value of each component fairly, we implement a game-theoretic analysis using Shapley values (Shapley 1953) across all $2^3 = 8$ possible component coalitions. Over the uniform flow baseline (CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2256 (85.8% of total gain) to

Table 4 Downstream CPC performance on the 25 held-out target cities under different decay calibration strategies using real Meta prior data (RQ1).

Calibration Strategy	**Data Source**	**Bin Configuration**	**Downstream CPC (**
`Meta MDM Prior (3-Bin)` (Survey-Free Prior)	Meta Movement Maps	3 Physical Bins	0.7080
`Third-Party Baseline (3-Bin)` (Survey-Based)	Third-Party Mobility Data	3 Physical Bins	0.7348
`Third-Party Oracle (20-Bin)` (Survey-Based)	Third-Party Mobility Data	20 Equal-Width Bins	0.7403
`DeepGravity (Zero-Shot)` (Supervised Baseline)	Multi-Source Features	Neural MLP	0.7727

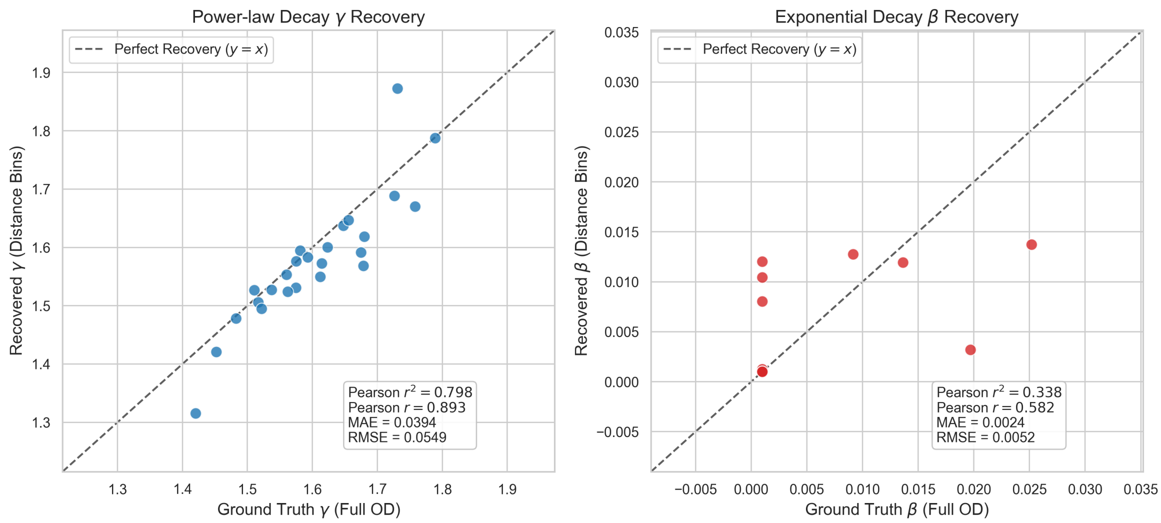

Fig. 6 Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent γ (left) and exponential decay rate β (right) show consistency in spatial decay structure (Pearson $r = 0.893$ for γ and $r = 0.582$ for β).

distance decay $f(d)$, 0.0207 (7.9% of total gain) to attractiveness A_j, and 0.0167 (6.4% of total gain) to outflow production O_i. This indicates that while distance decay is the dominant driver of spatial interaction, attraction and production features provide distinct and critical predictive improvements, with attraction having a slightly higher marginal importance. Crucially, this overwhelming dominance of distance decay explains why the aggregate calibration of the decay function (investigated in RQ1) serves as the primary driver of zero-shot transfer success: getting the dominant mechanism right is mathematically sufficient to recover the bulk of the transferable signal.

5.4 Zero-Shot Transfer Evaluation (Answering RQ3)

We now present an evaluation of the proposed aggregate-calibrated model against the baselines to determine the urban morphological conditions under which this is most effective (RQ3).

5.4.1 Comparison with Baselines (RQ3)

The findings are summarized in Table 6 below.

The Survey-Free version of the proposed model achieves a mean CPC of 0.6893, while the Oracle O_i variant reaches CPC 0.7398. Under zero-shot settings, DeepGravity outperforms the proposed model (CPC 0.7529 vs 0.6893), which is expected as DeepGravity leverages a high-capacity neural network to model complex non-linear feature interactions directly from the pooled training dataset. However, the proposed approach maintains a structured, physically interpretable formulation with far fewer features and parameters. Traditional Tanner Gravity calibrated locally on the target cities reaches a mean CPC of 0.7411,

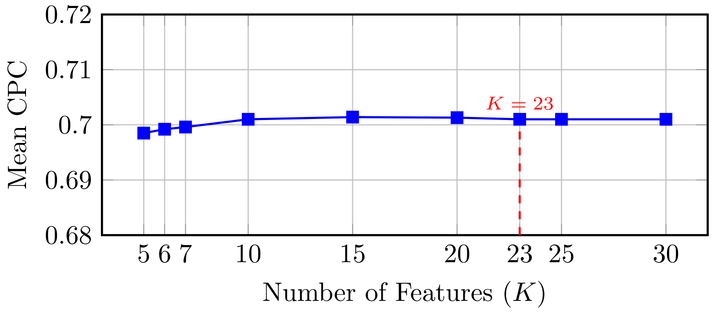

Fig. 7 SHAP feature count sweep over the 30-candidate feature set: zero-shot CPC first reaches 0.7010 at $K = 10$ and stabilizes in the range 0.7010–0.7014 through $K = 23$. The stability analysis retains $K = 23$ features (all passing both thresholds: $\mu_p \geq 0.005$, $S_p \geq 1.0$), achieving 23.3% dimensionality reduction without performance loss. The final deployed model uses the 6 core structural features (Section 3.4), confirmed by this sweep as the dominant contributors.

Table 5 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities). The full-model baseline (top row) uses a fixed national decay prior ‡ to isolate component contributions; this should not be confused with the aggregate-recovered decay used in the final proposed model.

$\mathbf{O}_i$	f(d)	$\mathbf{A}_j$	CPC	Factorial Contribution	Shapley Value
✓	✓‡	✓	0.6893	Baseline (all components, national prior decay)	–
×	✓‡	✓	0.6642	O_i removal: 3.6% loss	0.0167 (6.4%)
✓	×	✓	0.4489	f(d) removal: 34.9% loss	0.2256 (85.8%)
✓	✓‡	×	0.6571	A_j removal: 4.7% loss	0.0207 (7.9%)

‡In all ablation rows, $f(d)$ is frozen at the national prior $(\gamma_{\text{nat}}, \beta_{\text{nat}})$ to isolate the contribution of O_i and A_j. The full pipeline uses aggregate-recove

which serves as a locally-fit structural benchmark. The proposed model (Oracle O_i) closely approaches this locally-fit structural upper bound (0.7398 vs 0.7411) without using any target-city OD data.

5.4.2 Statistical Significance Testing

To assess the statistical reliability of our comparisons, we apply the Wilcoxon signed-rank test (Wilcoxon 1945) across the 25 held-out target cities under the 25/25 stratified split—a non-parametric paired test appropriate for our small, non-Gaussian sample size.

The proposed model (Oracle O_i) is practically equivalent to locally-calibrated Traditional Gravity: despite a statistically detectable difference (Wilcoxon $p = 0.0136$, Cohen's $d = -0.50$), the mean CPC gap is only 0.0013 (0.18% relative), which is negligible for mobility planning purposes—confirming that survey-free zero-shot transfer achieves practically the same predictive fidelity as a model fitted directly on the target city using local OD matrices. It strongly and significantly outperforms the Radiation Model baseline (ΔCPC $= +0.2547$, Wilcoxon $p = 5.96 \times 10^{-8}$, Cohen's $d = +6.85$). The gap versus DeepGravity (Oracle O_i) is statistically significant (ΔCPC $= -0.0127$, Wilcoxon $p = 5.56 \times 10^{-4}$, Cohen's $d = -0.82$), reflecting the expected advantage of a high-capacity neural model over a parsimonious, interpretable structure.

5.4.3 Urban Morphology Analysis (RQ3)

To address RQ3 and evaluate how urban morphological features affect zero-shot transfer performance, we analyze performance across the four morphological groups defined in Section 4.4. The transfer CPC results are shown in Table 7.

The structured physical formulation exhibits its strongest performance in Coastal environments (CPC 0.713), representing a 9.5% improvement over Dense / Transit environments (CPC 0.651). Crucially, the performance gap between the physics-constrained gravity model and the unconstrained neural DeepGravity model narrows to just 5.9% in Coastal cities (0.713 vs 0.758) compared to a much wider 10.3% gap in Dense /

Table 6 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Zero-Shot Transfer (No target-city travel survey data)				
Proposed Model (Survey-Free)	0.6893	0.0417	Decomposed	6 (selected)
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27 (all)
Group 2: Oracle Outflow Transfer (Zero-shot transfer with true outflows $O_i^{\mathbf{true}}$)				
Proposed Model (Oracle O_i)	0.7398	0.0287	Decomposed	6 (selected)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27 (all)
Group 3: Locally-Calibrated Baselines (Fitted directly on target cities)				
Naive Population Baseline	0.6758	0.0384	Baseline	0
Traditional Tanner Gravity	0.7411	0.0289	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.4851	0.0516	Param-free	0

Fairness note: Both the Proposed Model and DeepGravity (Groups 1–2) are trained exclusively on the same 25 source cities and evaluated on the same 25 held-out target cities. The feature count difference (6 vs. 27) reflects each model's own design: the proposed model uses 6 core structural features for zero-shot transfer, while DeepGravity uses its full feature set as defined in the original work (Simini et al. 2021). Note on DeepGravity CPC values: the Zero-Shot E2E mean CPC of 0.7529 (Group 1) and Oracle mean CPC of 0.7526 (Group 2) are evaluated under the 25/25 split. These differ from the 0.7727 in Table 4 (which uses a different calibration subset).

Table 7 Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	Proposed CPC	DeepGravity CPC	Rel. Gap[†] (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.691	0.760	9.1%
Polycentric	7	0.704	0.759	7.3%
Coastal	3	0.713	0.758	5.9%

[†]Relative gap = $(\mathrm{CPC}_{\mathrm{DG}} - \mathrm{CPC}_{\mathrm{Proposed}})/\mathrm{CPC}_{\mathrm{DG}} \times 100\%$.

Transit cities (0.651 vs 0.726). This significant narrowing of the performance gap demonstrates that in physically fragmented terrains (such as coastlines), physical distance friction acts as a dominant, highly constrained predictor of commuting flow, allowing the structured gravity formulation to perform close to the neural end-to-end model. In contrast, dense transit environments decouple travel from simple geometric distance, allowing DeepGravity's high-capacity feature modeling to widen its lead.

6 Discussion

6.1 What Aggregate Distributions Reveal About Distance Decay (RQ1)

The empirical results answer RQ1 by demonstrating that aggregate trip-length histograms preserve sufficient information to recover city-specific distance-decay parameters. The high correlation ($r = 0.893$ for the power-law exponent γ) and the negligible downstream CPC gap (0.18%) indicate that the spatial friction of distance is structural and can be captured from marginal distributions.

Physically, this is because a trip-length histogram represents a projection of a city's spatial interaction dynamics onto a single distance dimension. While individual origin-destination (OD) flows are highly sensitive to local land-use fluctuations, the rate at which travel probability decays with distance is a systemic urban property. Historically, calibrating these decay parameters required complete, individual-level OD travel surveys or invasive GPS/cellular telemetry, which are often legally restricted under privacy frameworks like GDPR. Our findings reveal that the individual movement probability data collected by tech platforms such as Meta can be aggregated into anonymous, population-level trip-length distributions to fully replace invasive individual trajectories for calibration.

From an information-theoretic perspective, although the spatial aggregation process discards the specific joint probability $P(i, j)$ of individual zone-to-zone transitions, it retains the marginal distribution over distance $p(d)$. Because the underlying physical friction of distance governs the shape of this marginal distribution, Maximum Likelihood Estimation (MLE) can robustly reconstruct the city-specific decay curve. This proves that high-fidelity gravity calibration does not require tracking individual OD pairs. By showing that aggregate, privacy-preserving probability distributions are practically equivalent to complete OD matrices for decay recovery, this study opens up new pathways for deploying urban mobility models in data-scarce or highly regulated regions.

6.2 Distance Decay as the Dominant Transferable Mechanism (RQ2)

Answering RQ2, the Shapley value analysis reveals that distance decay is the dominant transferable component of spatial interaction, explaining 85.8% of zero-shot predictive capacity over a uniform baseline. This dominance yields an important practical corollary: the zero-shot transfer pipeline exhibits high tolerance to prediction errors in other components. Because origin production (O_i) and destination attraction (A_j) operate within the strict physical boundaries set by distance decay, localized errors in GBDT outflow predictions or attraction heuristics are heavily suppressed during multiplicative gravity fusion. This explains why using a training-free attraction heuristic yields performance virtually equivalent to a trained model, and why the pipeline maintains robust transferability without requiring target-city OD data.

6.3 Scientific and Morphological Implications (RQ3)

The experimental evidence across 50 US metropolitan areas carries broader implications for how transferable urban mobility should be modeled, particularly under varying morphological conditions (RQ3).

Morphological scope and physical constraints. Our findings show that the efficacy of aggregate-calibrated gravity models is strongly governed by physical geography. In Coastal environments, where water bodies and topographic barriers funnel travel into constrained corridors, physical distance remains a dominant predictor of commuting flows. In these environments, the physics-constrained gravity model performs closest to the high-capacity neural baseline (DeepGravity), narrowing the performance gap to just 5.9%. Conversely, in Dense/Transit environments, multi-modal transit networks and dense spatial opportunities decouple travel from simple geometric distance, allowing neural models to exploit non-linear features and widen their lead (10.3% gap). This establishes the morphological boundary conditions of the gravity formulation: physical representations are most powerful where urban geography naturally limits spatial choice.

Shifting the zero-shot modeling paradigm. Traditionally, achieving zero-shot transfer has been treated as a representation learning problem, requiring neural networks to align complex, high-dimensional feature spaces across cities. Our results suggest an alternative paradigm: because distance decay is the dominant transferable mechanism, zero-shot transfer is primarily a calibration problem of the decay function. By proving that this calibration can be performed using only target-city aggregate histograms, this study shows that simple, interpretable physical models can achieve high transfer performance without the data-dependency and black-box nature of deep learning alternatives.

6.4 Practical and Data Privacy Implications

The feasibility of calibrating transferable gravity models from aggregate data offers significant practical advantages:

Privacy-Preserving Deployment : Because trip-length distributions do not contain individual OD pairs, they are significantly easier to release under strict data-privacy regulations (e.g., GDPR), enabling calibration in data-scarce or highly regulated regions.

Model Auditability : Unlike black-box neural models, the decomposed gravity formulation allows planners to independently modify attraction, production, or decay for policy simulations and "what-if" planning scenarios.

6.5 Limitations

1. **Separability Assumption**: The model assumes approximate separability between production, attraction, and decay. In cities with highly integrated multi-modal transit networks or strong spatial interaction effects, this independence assumption may be violated, leading to localized prediction errors.
2. **Geographic Scope**: The evaluation is limited to 50 US metropolitan areas. Cities in other regions may exhibit substantially different decay behavior: compact Asian megacities (e.g., Tokyo, Shanghai) typically show steeper short-range decay due to high-density mixed land use, while informal settlements in developing countries may exhibit weaker distance friction due to constrained road networks and non-motorized trip modes. Future work should validate the aggregate calibration approach on multi-continental datasets to assess cross-regional transferability.
3. **Demographic Representation**: Input datasets like cellular location logs underrepresent elderly, low-income, and smartphone-free populations, introducing potential demographic bias in the output OD flows (Gallotti et al. 2024).

7 Conclusions

This study investigated the information content of aggregate travel distributions for urban mobility modeling, leading to a central scientific discovery: **aggregate mobility distributions contain sufficient information to recover transferable distance-decay functions, and because distance decay is the dominant transferable mechanism, this enables accurate zero-shot mobility modeling without access to individual OD trajectories.**

Specifically, we have shown that:

- Anonymized trip-length histograms successfully recover city-specific distance-decay parameters with high fidelity ($r = 0.893$ against OD-fitted parameters).
- Distance decay is the dominant transferable component of spatial interaction, explaining 85.8% of cross-city predictive capacity.
- Consequently, using aggregate-recovered decay in downstream flow models achieves zero-shot predictive performance comparable to utilizing full OD matrices, with the relative gap to neural baselines shrinking to 5.9% in physically fragmented coastal environments.
- Under the 25/25 stratified evaluation, the proposed model is practically equivalent to locally-calibrated Traditional Gravity—despite a statistically detectable difference (Wilcoxon $p = 0.0136$, Cohen's $d = -0.50$), the mean CPC gap is only 0.0013 (0.18% relative)—while requiring no target-city OD data, demonstrating robust zero-shot generalizability across diverse US metropolitan areas.

By demonstrating that aggregate mobility data can effectively replace individual trajectories for calibration, this work establishes a privacy-preserving and structurally transparent pathway for zero-shot transfer in urban planning.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

Parameters (γ, β) are estimated via MLE from the aggregate trip-length histogram using L-BFGS-B optimization in log-space with 10 random restarts to avoid local optima. Full implementation details are available in the code repository.

A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

Bin Sensitivity Analysis Sweeping the distance bins $K \in \{3, 5, 10, 20\}$ shows that downstream prediction performance stabilizes at a high level. Specifically, the mean CPC across target cities is 0.7346 for $K = 3$, 0.7367 for $K = 5$, 0.7391 for $K = 10$, and 0.7398 for $K = 20$ (compared to the full OD ground truth baseline of 0.7411).

This indicates that even extremely coarse spatial data (e.g., 5 to 10 equal-width bins) is sufficient to robustly capture the decay profile and predict commuting flows.

Missing Attractiveness Data (A_j) Randomly removing 10%, 20%, and 50% of target-city destination attractions A_j yields highly stable decay recovery (e.g., $\gamma_{\mathrm{MAE}} = 0.0452$ under 20% missing data vs. 0.0394 baseline). This validates the robustness of aggregate MLE against localized data gaps.

CBD Collapse (Extreme A_j Distortion) Disrupting the attraction A_j of the top 10% highest attractiveness zones (CBD collapse) still allows accurate parameter recovery under two scenarios: replacing affected zones with the city-wide mean attraction ($\gamma_{\mathrm{MAE}} = 0.0541$) and scaling them down by 80% ($\gamma_{\mathrm{MAE}} = 0.0600$). This proves that spatial decay friction can be recovered even when key destination hubs are heavily distorted.

A.3 Preprocessing and Normalization

Data preprocessing details are summarized in Table 8.

Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of zero-shot production transfer across two dimensions: training-set scale and prediction noise.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of $n_{\mathrm{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from $n_{\mathrm{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Even under substantial injected noise (40%), the target CPC degrades only marginally (from 0.6893 to 0.6623). This demonstrates that production-constraint normalization effectively dampens localized estimation noise.

Appendix C. Feature Selection Sensitivity

C.1 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 6, 7, 10, 15, 20, 23, 25, 30\}$) shows that predictive performance plateaus between 10 and 23 features (CPC: 0.7010). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators, with additional features providing marginal utility.

Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 10.

References

Alis C, Legara EFT, Monterola C (2021) Generalized radiation model for human migration. Scientific Reports 11(1):22707. https://doi.org/10.1038/s41598-021-02109-1

Atwal KS, Anderson T, Pfoser D, et al (2025) Commuting flow prediction using OpenStreetMap data. Computational Urban Science 5(1):2. https://doi.org/10.1007/s43762-025-00161-5

Barbosa H, Barthelemy M, Ghoshal G, et al (2018) Human mobility: Models and applications. Physics Reports 734:1–74. https://doi.org/10.1016/j.physrep.2018.01.001

Brockmann D, Hufnagel L, Geisel T (2006) The scaling laws of human travel. Nature 439(7075):462–465. https://doi.org/10.1038/nature04292

Gallotti R, Maniscalco D, Barthelemy M, et al (2024) Distorted insights from human mobility data. Communications Physics 7(1):1–10. https://doi.org/10.1038/s42005-024-01909-x

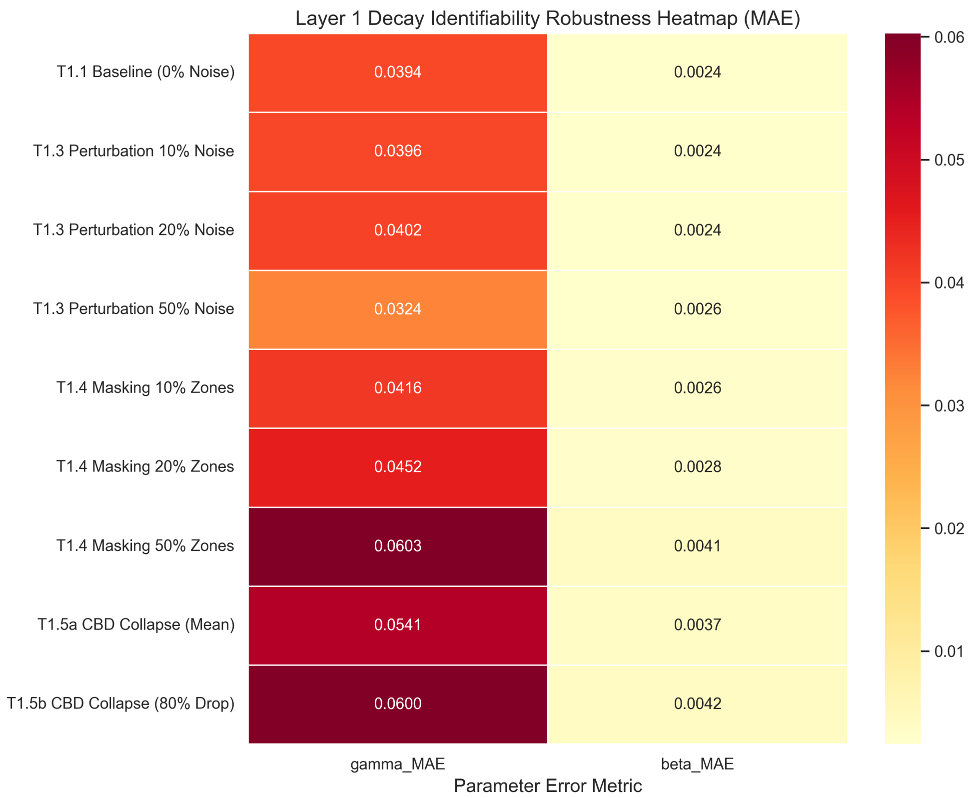

Fig. 8 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

Table 8 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	**Treatment**	**Rationale**
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

Table 9 GBDT Model Hyperparameter Configurations

Hyperparameter	**Outflow Model (O_i)**	**Attraction Model (A_j, ablation only)**
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	6	6
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

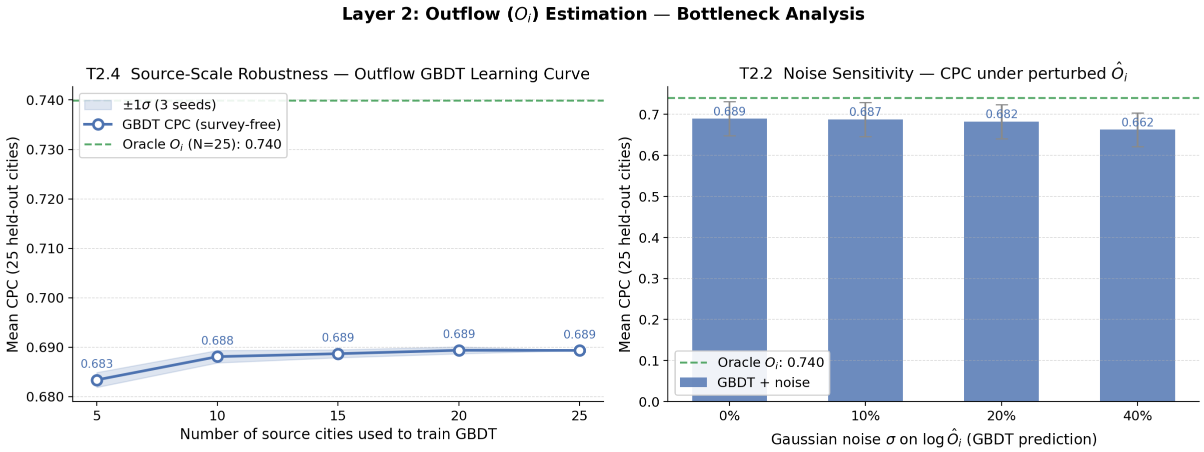

Fig. 9 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

Table 10 List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations
SDF-ZTF	Separable Decomposed Formulation for Zero-Shot Transferable Flows		

González MC, Hidalgo CA, Barabási AL (2008) Understanding individual human mobility patterns. Nature 453(7196):779–782. https://doi.org/10.1038/nature06958

Jin G, Wu Y, Wang M, et al (2022) Selective cross-city transfer learning for traffic prediction via source city region re-weighting. In: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. Association for Computing Machinery, pp 729–739, https://doi.org/10.1145/3534678.3539345

Lenormand M, Huet S, Gargiulo F, et al (2012) A universal model of commuting networks. PLoS One 7(10):e45985. https://doi.org/10.1371/journal.pone.0045985

Lenormand M, Bassolas A, Ramasco J (2016) Systematic comparison of trip distribution laws and models. Journal of Transport Geography 51:158–169. https://doi.org/10.1016/j.jtrangeo.2016.02.003

Lundberg SM, Lee SI (2017) A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30:4765–4774. https://doi.org/10.48550/arXiv.1705.07874

Mai G, Cundy C, Choi K, et al (2022) Towards a foundation model for geospatial artificial intelligence. ACM Transactions on Intelligent Systems and Technology 13(6):1–25. https://doi.org/10.1145/3557915.3561043

Mendieta M, Han B, Shi X, et al (2023) Towards geospatial foundation models via continual pretraining. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp 16781–16790, https://doi.org/10.1109/ICCV53024.2023.01529

Meta (2026) Movement distribution maps. https://ai.meta.com/ai-for-good/datasets/movement-distribution-maps/, aI for Good Datasets. Accessed: 27 June 2026

Pappalardo L, Manley E, Sekara V, et al (2023) Future directions in human mobility science. Nature Computational Science 3:588–600. https://doi.org/10.1038/s43588-023-00469-4

Rong C, Feng J, Ding J, et al (2023) GODDAG: Generating origin-destination flow for new cities via domain adversarial training. IEEE Transactions on Knowledge and Data Engineering 35(10):10048–10057. https://doi.org/10.1109/TKDE.2022.3218526

Shapley LS (1953) A value for n-person games. In: Contributions to the Theory of Games, vol 2. Princeton University Press, p 307–317

Simini F, Gonzalez M, Maritan A, et al (2012) A universal model for mobility and migration patterns. Nature 484:96–100. https://doi.org/10.1038/nature10856

Simini F, Barlacchi G, Luca M, et al (2021) A deep gravity model for mobility flows generation. Nature Communications 12(1):5647. https://doi.org/10.1038/s41467-021-26752-6

Song C, Qu Z, Blumm N, et al (2010) Limits of predictability in human mobility. Science 327(5968):1018–1021. https://doi.org/10.1126/science.1177170

Stouffer SA (1940) Intervening opportunities: A theory relating mobility and distance. American Sociological Review 5(6):845–867. https://doi.org/10.2307/2084520

Tanner J (1961) Factors affecting the amount of travel. Road Research Technical Paper 51, Road Research Laboratory, London

Tatem AJ (2017) WorldPop, open data for spatial demography. Scientific Data 4:170004. https://doi.org/10.1038/sdata.2017.4, worldPop gridded population datasets available at https://www.worldpop.org/

Wang L, Geng X, Ma X, et al (2018) Crowd flow prediction by deep spatio-temporal transfer learning. In: Proceedings of the 27th International Joint Conference on Artificial Intelligence, pp 1341–1347, https://doi.org/10.24963/ijcai.2018/186

Wang S, Cao J, Yu PS (2021) Deep learning based origin-destination prediction via contextual information fusion. Multimedia Tools and Applications 80(20):30427–30449. https://doi.org/10.1007/s11042-020-09355-6

Wang X, Jin K, Sun L, et al (2022) When transfer learning meets cross-city urban flow prediction: Spatio-temporal adaptation matters. In: Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22), pp 2219–2225, https://doi.org/10.24963/ijcai.2022/308

Wilcoxon F (1945) Individual comparisons by ranking methods. Biometrics Bulletin 1(6):80–83. https://doi.org/10.2307/3001968

Wilson A (1971) A family of spatial interaction models, and associated developments. Environment and Planning A 3(1):1–32. https://doi.org/10.1068/a030001

Yao H, Liu Y, Wei Y, et al (2019) Learning from multiple cities: A meta-learning approach for spatial-temporal prediction. In: Proceedings of the Web Conference 2019, pp 2123–2129, https://doi.org/10.1145/3308558.3313577